Customer Analytics Data Warehouse Project

SQL-Based Customer Intelligence & Business KPI Reporting

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Customer Analytics Reporting System

Solving Real Business Problems Using SQL & Data Warehousing

Project Objective

Businesses collect massive amounts of transactional data but struggle to convert it into actionable insights. This project transforms raw customer and sales data into business intelligence metrics and decision-making tools.

Critical Problems Addressed

- Customer churn and weak retention
- Poor customer segmentation
- Ineffective marketing campaigns
- Limited visibility into customer value
- Low engagement detection
- Revenue performance analysis

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Business Challenges Addressed

No Visibility into High-Value Customers

Business Problem

The company could not identify which customers generated the highest long-term revenue, leading to inefficient marketing spend and missed retention opportunities.

SQL Solution

Built customer segmentation logic to identify VIP customers, regular customers, and new customers using sales thresholds and engagement metrics.

Improved Loyalty Strategies

Targeted retention programs for high-value customers

Increased Customer Lifetime Value

Better understanding of long-term revenue potential

Enhanced Retention

Proactive engagement with at-risk valuable customers

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Customer Churn Detection

Early Warning System for Revenue Loss

The Challenge

Customers stopped purchasing before the business noticed, resulting in significant revenue loss and missed win-back opportunities.

Technical Implementation

Created recency-based churn analysis using DATEDIFF to calculate time since last purchase and flag at-risk customers.

1

Early Detection

Identify customers likely to churn before they leave

2

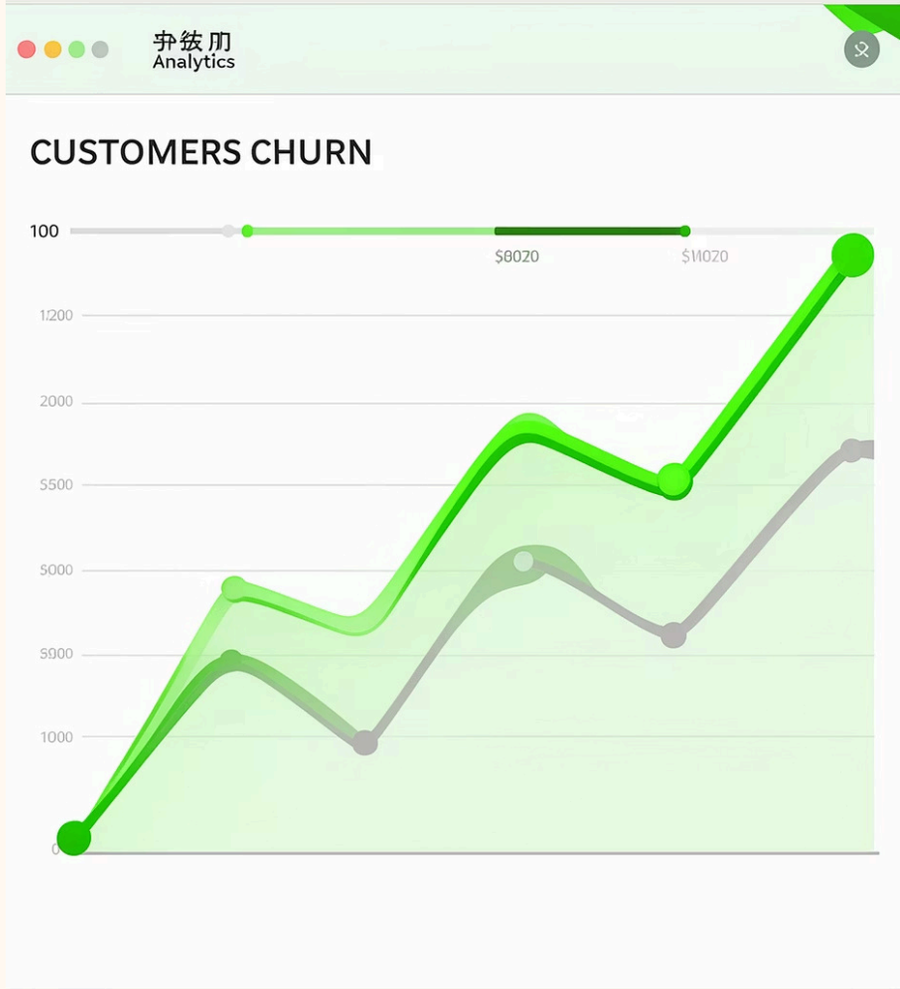
Win-Back Campaigns

Deploy targeted retention strategies at optimal timing

3

Reduced Revenue Loss

Minimize customer attrition through proactive intervention



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Low Engagement Customer Analysis

Targeting One-Time Buyers

Business Problem

Many customers placed only one order and never returned, representing significant lost revenue and poor conversion rates.

SQL Solution

WHERE total_orders = 1 filtering logic identifies low-engagement customers for targeted retention campaigns.

First-Repeat Discounts

Incentivize second purchase with strategic discount offers

Onboarding Campaigns

Guide new customers through product education

Educational Marketing

Build product knowledge and brand loyalty

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Customer Activity Monitoring

Comprehensive Behavior Tracking

Business Problem

The company lacked visibility into customer purchasing behavior, making it impossible to predict needs or identify engagement patterns.

Solution

Created comprehensive KPIs for proactive customer relationship management.

Total Orders

Track complete purchase history and frequency patterns

Products Purchased

Analyze category preferences and buying trends

Monthly Spending

Monitor seasonal patterns and spending behavior

Customer Lifespan

Calculate relationship duration and engagement longevity

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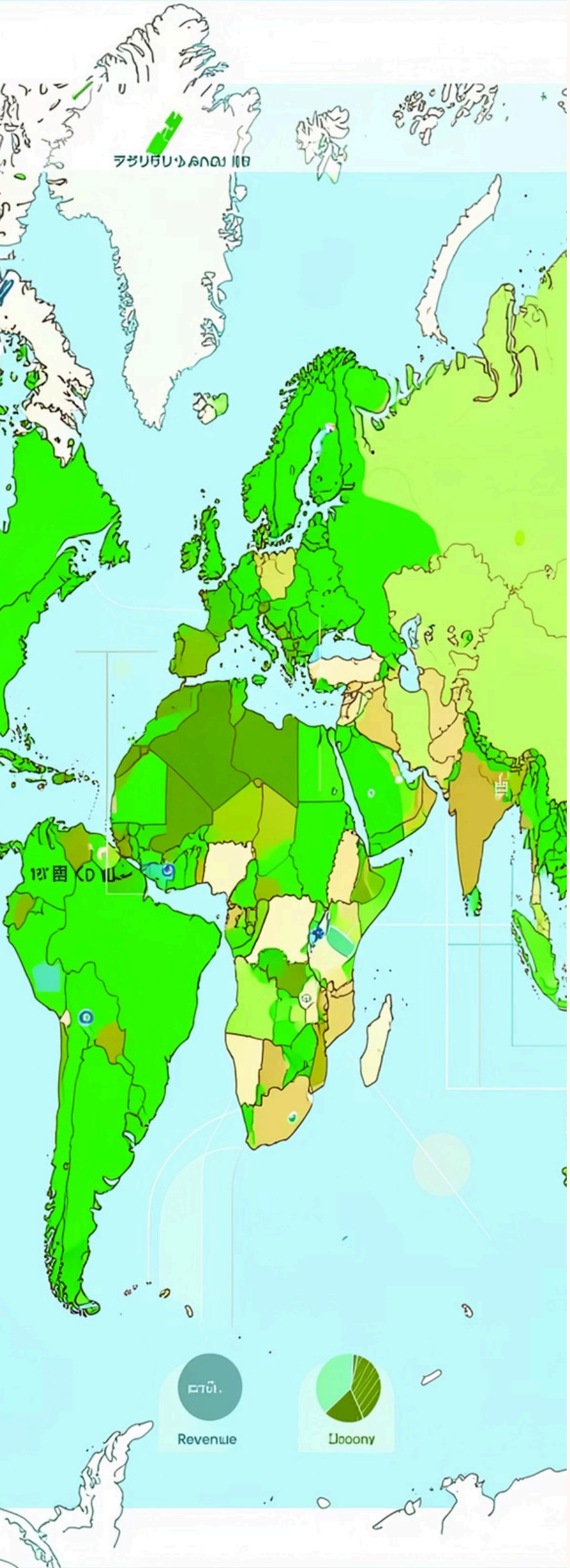

Revenue Intelligence Analysis

Demographic & Geographic Insights

Business Problem

Marketing teams did not know which customer groups generated the most revenue, resulting in inefficient advertising spend and poor targeting.

SQL Implementation

Built demographic and geographic revenue analysis using GROUP BY age_group and GROUP BY country queries.

Better Marketing ROI

Allocate budget to highest-performing segments

Smarter Advertising Spend

Optimize campaigns based on revenue data

Improved Audience Targeting

Reach the right customers with personalized messaging

Regional Sales Optimization

Adjust strategies by geographic performance

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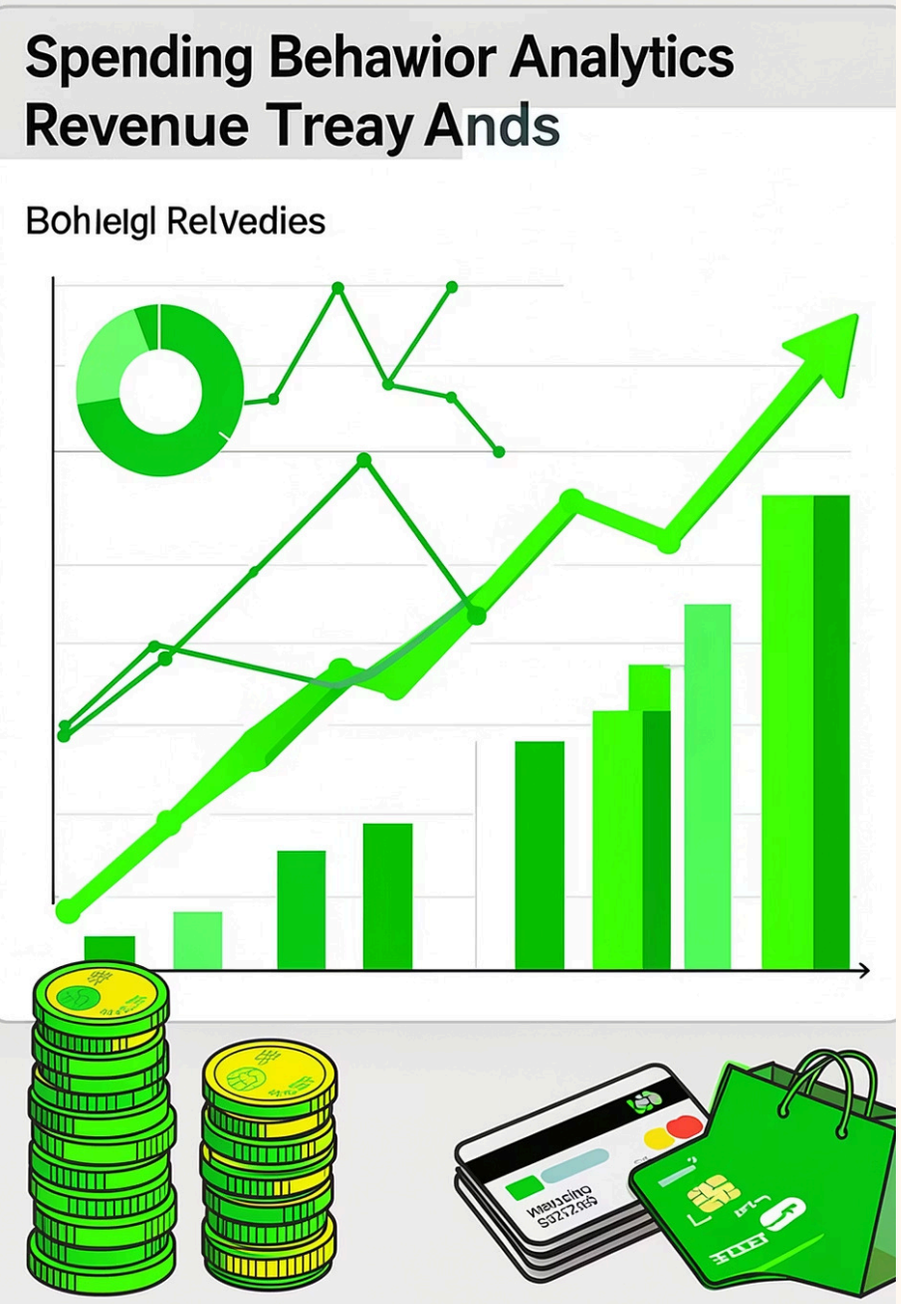

Spending Behavior Analytics

Understanding Customer Economics

Business Problem

The company lacked insight into customer spending patterns, preventing effective upselling and cross-selling strategies.

Solution

Created comprehensive KPIs including average order value, average monthly spend, and customer lifetime value calculations.

01

Average Order Value Analysis

Identify upselling opportunities in each transaction

02

Monthly Spend Tracking

Monitor recurring revenue patterns and predict future spend

03

Customer Lifetime Value

Calculate long-term revenue potential for strategic planning

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Customer Intelligence Framework

Segmentation Strategy

Business Problem

Marketing campaigns were too broad and ineffective, treating all customers the same regardless of value or behavior.

SQL Implementation

Implemented customer segmentation logic with clear thresholds.

VIP Customers

WHEN total_sales > 5000 AND lifespan_months >= 12 THEN 'VIP'

Regular Customers

Medium-value long-term customers requiring consistent engagement

New Customers

Recently acquired customers in onboarding phase

Personalized Campaigns

Tailor messaging to each segment's needs and behaviors

Better Conversion Rates

Relevant offers increase purchase likelihood

Higher Customer Satisfaction

Appropriate communication builds stronger relationships

Improved Retention Strategy

Proactive engagement prevents churn across all segments

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Technical Solution Design

Data Warehouse Architecture

Dimension Table

gold.dim_customers stores customer demographic information including age, location, acquisition date, and customer attributes.

Fact Table

gold.fact_sales stores transactional sales data with order details, product information, and revenue metrics.

Central Reporting Layer

Created reusable SQL view: gold.cus_report

Customer Behavior

Complete engagement tracking

Revenue Metrics

Comprehensive financial KPIs

Engagement KPIs

Recency and frequency analysis

Segmentation Logic

Automated customer classification

Business Impact: Unified all critical analytics into a single business-ready reporting system, eliminating the need for fragmented queries and manual data integration.

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